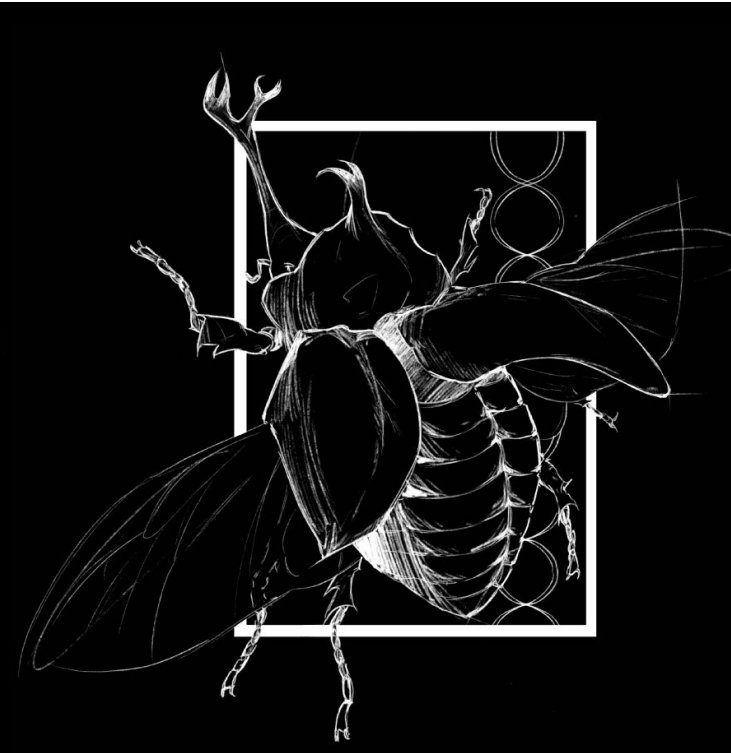




Stepwise Transition From Camouflage to Aposematism in Phasmatodea

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Background

Phasmatodea are arboreal insects that are found worldwide. As any living organism does, they find themselves under predation. Evolution has favored camouflage as their defensive strategy hence, their common name “stick bugs”. Most phasmatodea have slender bodies and limbs to mimic twigs, and they will often sway from side to side as a leaf does with a small breeze. Though this is the conventional depiction of stick bugs, they are diverse insects with a wide range of body shapes and colors. Some phasmatodea are called tree lobsters due to their thick limbs, dark red coloration, and resemblance to lobsters. More importantly, some species of phasmatodea have deviated from camouflage and rather use conspicuous signals (bright colors) as a warning to predators. The question is whether a deimatic display such as bright coloration on their hindwing precedes full body coloration (aposematism). Diemantic displays allow species to remain camouflaged (a.k.a. cryptic) until movement of their hindwings reveals bright colors and startles predators. This state might be a necessary development before evolving to full aposematism.

Study Design & AI

We defined three possible states for 120 of the species found in our tree.
Cryptic (0) = Resemblance to foliage, branches, or tree barks and ranged from colors such as light green to dark reds and browns.
Deimatic (1) = Possession of hindwings of any size with conspicuous signals such as a uniform bright color (commonly pink) or spots.
Aposematism (3) = Full body coloration that wouldn't be used for camouflage such as bright limbs or contrasting colors (e.g. black and red).
Each of these states were mapped onto our phylogeny in order to visually observe whether evolution of diemantic or aposematic displays were independent occurrences or demonstrated clustering as seen in Figure 1.
Claude AI supported our data collection by finding sources with descriptions of the species in our trees. Claude AI also aided in the production of R code for our figure 1 and analysis.

Model Fitting

To determine which evolutionary model best fit the data, we used a Markov model. This type of model is best for discrete traits because it determines which states are more stable according to specific rates such as those seen in figure 2. To observe this, two matrices were made and compared with one another. One Q matrix was constructed where transitions between cryptic and aposematic were not allowed (4 parameters). This model assumed that diemantic display is a necessary development before full body coloration, as the hypothesis states. The second model was unrestricted, so transitions between cryptic and aposematic states were allowed (6 parameters). The log likelihood of both models was calculated to determine how well the data fit into each model. The AIC and delta AIC values were used to compare both models with each other and determine which was a better representation of the character states.
Then, we simulated our dataset with a parametric bootstrap method that uses the parameters from observed data to simulate new datasets and determine whether the observed results were significant or due to chance.

Results

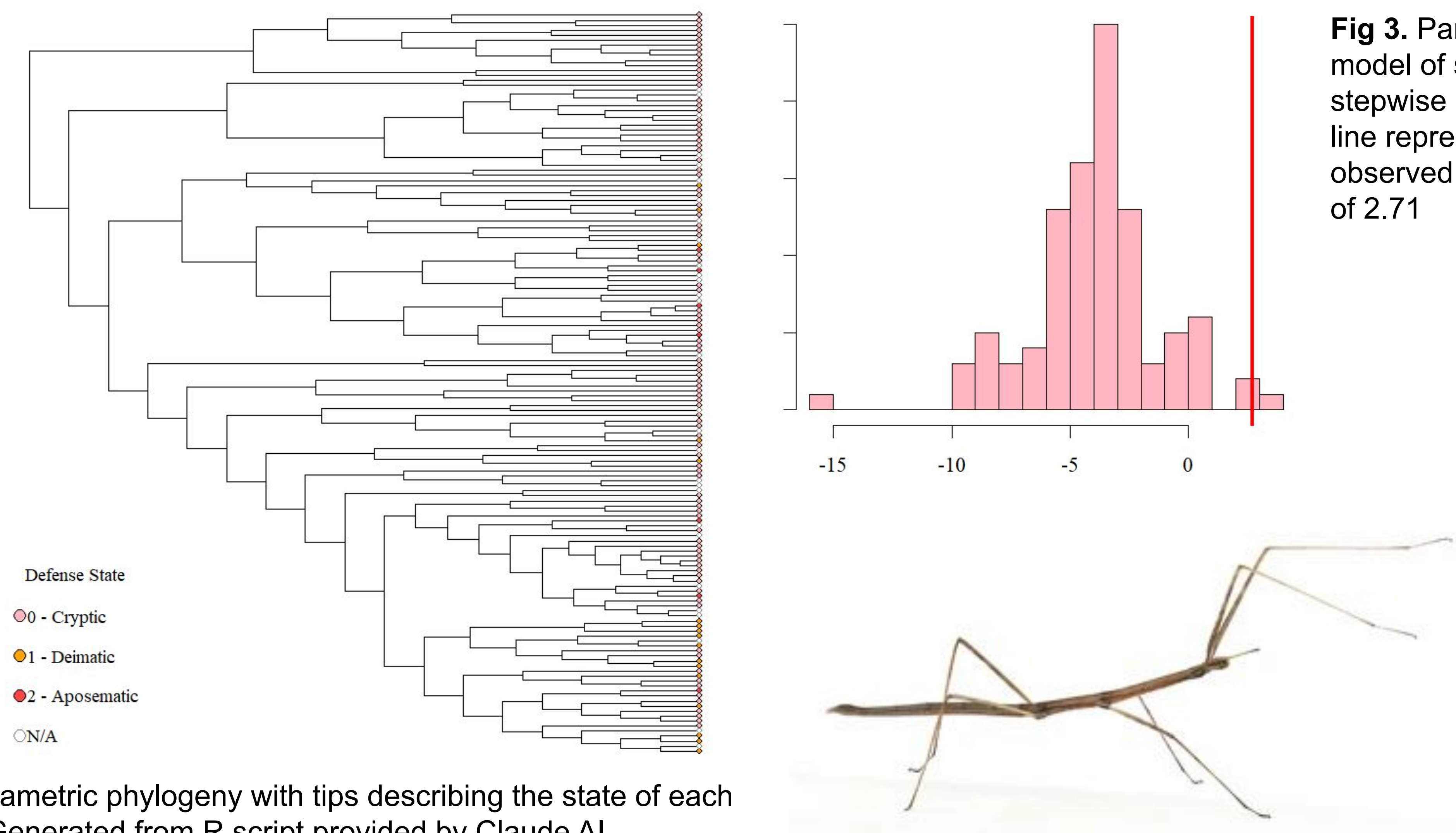


Fig. 1 Ultrametric phylogeny with tips describing the state of each species. Generated from R script provided by Claude AI

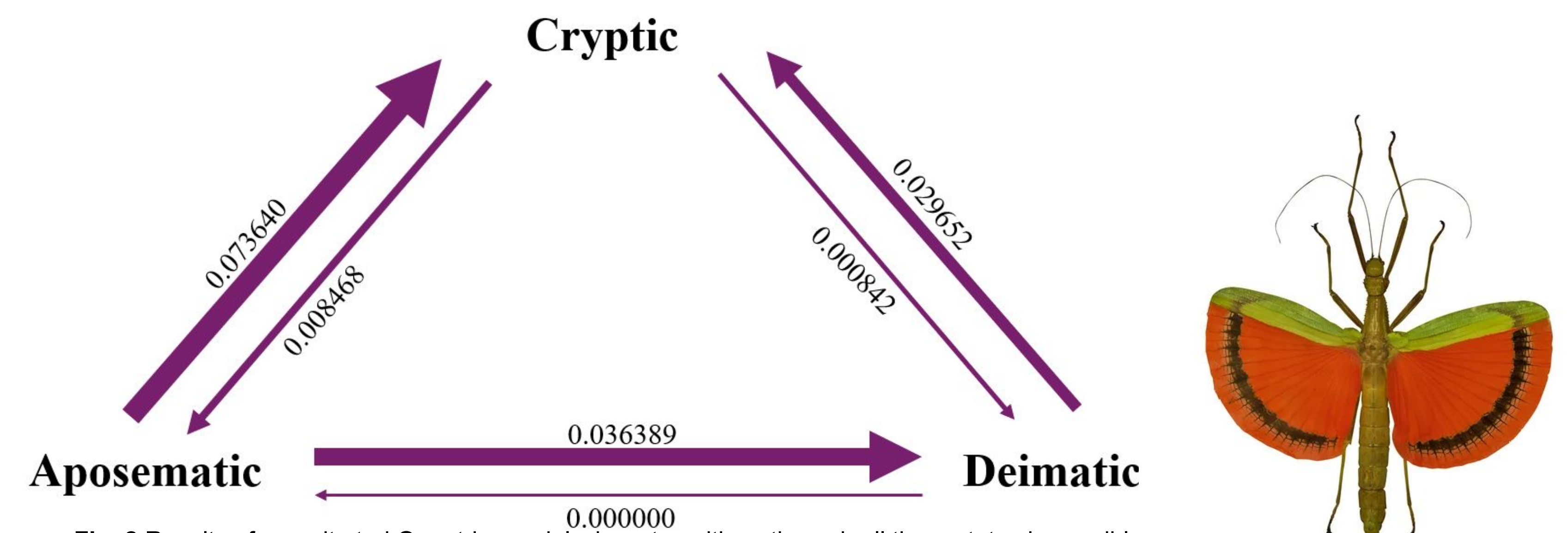
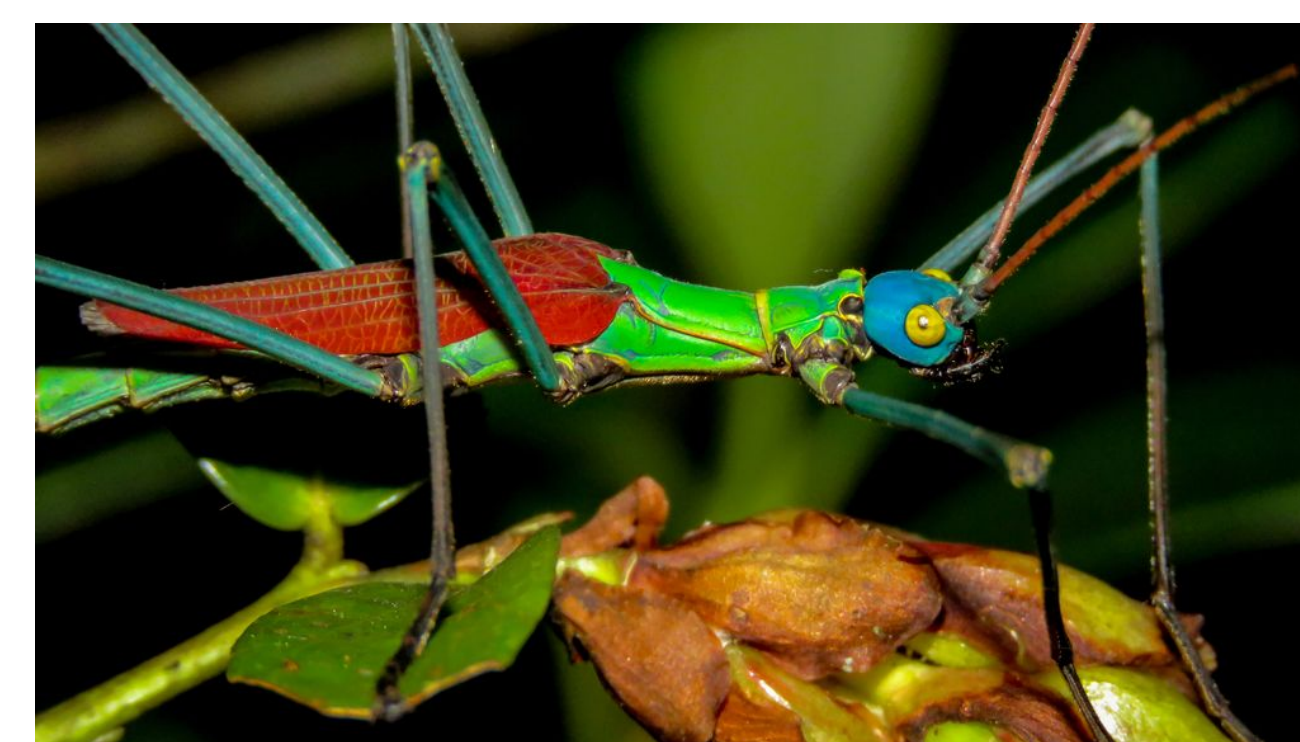


Fig. 2 Results of unrestricted Q matrix model where transitions through all three states is possible



Discussion

The restricted model yielded a log likelihood of -68.707119 while the unrestricted model yielded a log likelihood of -65.354567. These values indicate that the unrestricted model is more likely to be accurate, though it isn't a large difference. The AIC value for the unrestricted model was 142.7091 with a p value of 0.03499493. These values suggest that a transition to a deimatic state is not necessary to achieve full aposematism as our hypothesis stated. Rather, the values as seen on figure 2, demonstrate that a transition from cryptic to a deimatic state is rare (0.000842), while reversible transitions such as aposematic to cryptic or aposematic to deimatic happen more frequently. Notably, the delta AIC value for the restricted model yielded 2.705104. This isn't much higher than the restricted model and might suggest that our hypothesis is not completely out of the question. Additionally, progression from deimatic to aposematic yielded 0.0000 meaning it's completely absent. It could be possible that rather than using hindwing coloration as a defense strategy, it is used as a mating strategy. Furthermore, our delta AIC of 2.705104 fell outside of the range of the simulated stepwise model. The simulated model yielded a mean delta AIC of -3.949, 95% CI [-9.823, 1.853]. This provides further evidence that a deimatic stage is not necessary since our observed value falls outside of the expected range if our hypothesis were true.

While the data contained 120 species of Phasmatodea, there are over 3,000 recorded species worldwide that could be used to repeat this analysis. Bias due to a small sample size is a risk, however future research could factor that into data collection and analysis. Additionally, reliable descriptions were more difficult to find for some species than others, especially their habitat descriptions used to properly determine crypticism.

Conclusions & Acknowledgements

In conclusion, our analysis suggests that deimatic displays are not used as a stepping stone toward aposematic defensive strategies in Phasmatodea. While the delta AIC value for the restricted Q matrix was very low, the results do not support our hypothesis and might suggest that deimatic displays are not defensive.

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